JAVA PROGRAMMING PRACTICES

Bank Account Management System

Basic Information

Week	Week 1–2
Source file	BankAccountDemo.java
Concepts	Classes, objects, constructors, inheritance, methods, console input.

Objective / Aim

To implement bank account management system using the concepts listed in the R23 Java Programming Practices syllabus.

System Requirements

JDK (Java Development Kit) and a text editor or Java IDE.

Algorithm / Logic Description

1. Define the Account class and its account operations.
2. Extend Account for savings and current accounts.
3. Create account objects, perform deposit and withdrawal, and display details.

UML / Program Structure

Source Code

```
import java.util.Scanner;
class Account
{
    int accountNumber;
    double balance;
    Account(int accountNumber, double balance)
    {
        this.accountNumber = accountNumber;
        this.balance = balance;
    }
    void deposit(double amount)
    {
        if (amount > 0)
        {
            balance += amount;
            System.out.println("Deposited: " + amount);
        }
        else System.out.println("Invalid deposit amount");
    }
    void withdraw(double amount)
    {
        if (amount > 0 && amount <= balance)
        {
            balance -= amount;
            System.out.println("Withdrawn: " + amount);
        }
        else System.out.println("Insufficient balance or invalid amount");
    }
    void display()
    {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Balance: " + balance);
    }
}
class SavingsAccount extends Account
{
    double interestRate;
    SavingsAccount(int number, double balance, double rate)
    {
        super(number, balance);
        interestRate = rate;
    }
```

Source Code (continued)

```
    void calculateInterest()
    {
        System.out.println("Interest: " + (balance * interestRate / 100));
    }
}
class CurrentAccount extends Account
{
    double overdraftLimit;
    CurrentAccount(int number, double balance, double limit)
    {
        super(number, balance);
        overdraftLimit = limit;
    }
    void displayLimit()
    {
        System.out.println("Overdraft Limit: " + overdraftLimit);
    }
}
public class BankAccountDemo
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        SavingsAccount savings = new SavingsAccount(101, 5000, 5);
        CurrentAccount current = new CurrentAccount(102, 8000, 2000);
        System.out.println("SAVINGS ACCOUNT");
        savings.display();
        System.out.print("Enter deposit amount: ");
        savings.deposit(sc.nextDouble());
        System.out.print("Enter withdrawal amount: ");
        savings.withdraw(sc.nextDouble());
        savings.calculateInterest();
        System.out.println("\nCURRENT ACCOUNT");
        current.display();
        current.displayLimit();
        sc.close();
    }
}
```

Compilation and Execution

```
javac BankAccountDemo.java
java BankAccountDemo
```

Output Screen

```
SAVINGS ACCOUNT
Account Number: 101
Balance: 5000.0
Enter deposit amount: Deposited: 1000.0
Enter withdrawal amount: Withdrawn: 500.0
Interest: 275.0

CURRENT ACCOUNT
Account Number: 102
Balance: 8000.0
Overdraft Limit: 2000.0
```

Result

The program demonstrates the listed syllabus concepts. The sample output is based on a compiled execution.